JAN 18 1991

SUPERSEDING
MIL-R-85690A(AS)
19 February 1985

MILITARY SPECIFICATION

RADIO RECEIVING SET AN/ARN-146

This specification is approved for use within the Naval Air Systems Command, Department of the Navy, and is available for use by all Departments and Agencies of the Department of Defense.

1. SCOPE

1.1 Scope. This specification establishes the design, performance, and acceptance requirements for the AN/ARN-146 Radio Receiving Set, herein referred to as the receiving set. The receiving set is to be an Airborne ASW Sonobuoy Receiver which receives Sonobuoy transmissions in the VHF band and provides homing and On-Top Position Indicator (OTPI) information when used in conjunction with a direction finder equipment.

1.2 Classification. This specification establishes the requirements for the following items as noted herein.

Nomenclature	Description
Receiver Set, Radio AN/ARN-146	The receiving set with Radio Set Control.
Receiver Set, R-2330/ARN-146	The receiving set without Radio Set Control.
Control, Radio Set C-11699/ARN-146	The receiving set Radio Set Control only.
Mounting Base, Electrical Equipment	MT-6485/ARN-146

Beneficial comments (recommendations, additions, deletions) and any pertinent data which may be of use in Improving this document should be addressed to: Commanding Officer, Naval Air Engineering Center, Systems Engineering and Standardization Department (SESD) Code 53, Lakehurst, NJ 08733-5100, by using the Standardization Document Improvement Proposal (DD Form 1426) appearing at the end of this document or by letter.

AMSC N/A FSC 5826

DISTRIBUTION STATEMENT A. Approved for public release; distribution is unlimited.

Source: https://assist.dla.mil -- Downloaded: 2026-10-26T04:51Z
Check the source to verify that this is the current version before use.

1.3 Associated equipment. This receiving set should operate with the associated equipment listed in 6.9.

2. APPLICABLE DOCUMENTS

2.1 Government documents.

2.1.1 Specification, standards, and handbooks. The following specifications, standards, and handbooks form a part of this document to the extent specified herein. Unless otherwise specified, the issues of these documents are those listed in that issue of the Department of Defense Index of Specifications and Standards (DoDISS) and supplement thereto, cited in the solicitation (see 6.2).

SPECIFICATIONS

MILITARY

MIL-B-5087	Bonding, Electrical, and Lightning Protection for Aerospace Systems
MIL-W-5088	Wiring Aerospace Vehicle
MIL-E-5400	Electronic Equipment, Aerospace, General Specification for
MIL-T-5422	Testing Environmental, Airborne Electronic and Associated Equipment
MIL-C-6781	Control Panel; Aircraft Equipment, Rack or Console Mounted
MIL-F-7179	Finishes, Coatings, and Sealants for the Protection of Aerospace Weapons Systems
MIL-M-7793	Meter, Time Totalizing, Miniature, Digital, 28VDC
MIL-I-8500	Interchangeability and Replaceability of Component Parts for Aerospace Vehicles
MIL-E-17555	Electronic and Electrical Equipment Accessories, and Provisioned Items (Repair Parts), Packaging of
MIL-M-18012	Markings for Aircrew Station Displays, Design and Configuration of
MIL-T-18303	Test Procedures; Preproduction, Acceptance and Life for Aircraft Electronic Equipment, Format for
MIL-N-18307	Nomenclature and Identification for Aeronautical Systems including joint Electronics type designated systems and associated support systems
MIL-S-19500	Semiconductor Devices, General Specification for
MIL-B-22356	Bathythermograph Transmitting Set, AN/SSQ-36

Source: https://assist.dla.mil -- Downloaded: 202610-26T04:51Z
Check the source to verify that this is the current version before use.

SPECIFICATIONS

MILITARY

MIL-C-25050	Colors, Aeronautical Lights and Lighting Equipment, General Requirements for
MIL-M-38510	Microcircuits, General Specification for
MIL-C-38999	Connector, Electrical, Circular, Miniature, High Density, Quick Disconnect (Bayonet, Threaded, and Breech Coupling), Environmental Resistant, Removable Crimp and Hermetic Solder Contacts, General Specification for
MIL-C-39012	Connectors, Coaxial, Radio Frequency, General Specification for
MIL-P-55110	Printed Wiring Boards, General Specification for
MIL-R-81355	Recorder, Bathythermograph Data RO-308/SSQ-36
MIL-S-81478	Sonobuoy AN/SSQ-57, Specification for
MIL-S-81487	Sonobuoy AN/SSQ-53, Specification for
MIL-C-81511	Connectors, Electrical, Circular, High Density, Quick Disconnect, Environment Resisting: And Accessories, General Specification for
MIL-R-81680	Receiver Radio R-1651/ARA
MIL-C-81774	Control Panel, Aircraft, General Requirements for
MIL-S-85020	Sonobuoy AN/SSQ-71
MIL-R-85689	Radio Receiving Set AN/ARR-84

STANDARDS

FEDERAL

FED-STD-595	Colors

MILITARY

MIL-STD-275	Printed Wiring for Electronic Equipment
MIL-STD-454	Standard General Requirements for Electronic Equipment
MIL-STD-461	Electromagnetic Emission and Susceptibility Requirements for the control of Electromagnetic Interference

Source: https://assist.dla.mil -- Downloaded: 2026-10-26T04:51Z
Check the source to verify that this is the current version before use.

STANDARDS

MILITARY

MIL-STD-462	Electromagnetic Interference Characteristics, Measurement of
MIL-STD-471	Maintainability/Verification/Demonstration/Evaluation
MIL-STD-481	Configuration Control - Engineering Changes, (Short Form), Deviations, and Waivers
MIL-STD-483	Configuration Management Practices for Systems, Equipment, Munitions, and Computer Programs
MIL-STD-704	Aircraft Electric Power Characteristics
MIL-STD-756	Reliability Modeling and Prediction
MIL-STD-781	Reliability Testing for Engineering Development, Qualification, and Production
MIL-STD-783	Legends for use in Aircrew Stations and on Airborne Equipment
MIL-STD-785	Reliability Program for System and Equipment Development and Production
MIL-STD-794	Parts and Equipment, Procedures for
MIL-STD-810	Environmental Test Methods and Engineering Guidelines
MIL-STD-882	System Safety Program Requirements
MIL-STD-883	Test Methods and Procedures for Microelectronics
MIL-STD-965	Parts Control Program
MIL-STD-1353	Electrical Connectors, Plug-in Sockets and Associated Hardware, Selection and use of
MIL-STD-1472	Human Engineering Design Criteria for Military Systems Equipment and Facilities
MIL-STD-1629	Procedures for Performing a Failure Mode and Effect Analysis for Shipboard Equipment
MIL-STD-1686	Electrostatic Discharge Control Program for Protection of Electrical and Electronic Parts, Assemblies, and Equipment (Excluding Electrically Initiated Explosive Devices) (Metric)

Source: https://assist.dla.mil -- Downloaded: 2026-10-26T04:51Z
Check the source to verify that this is the current version before use.

STANDARDS

MILITARY

MIL-STD-2000	Standard Requirements for Soldered Electrical and Electronic Assemblies
MIL-STD-2074	Failure Classification for Reliability Testing
MIL-STD-2076	Unit under Test Compatibility with Automatic Test Equipment, General Specification for
MIL-STD-2077	Test Program Sets, General Requirements for
MIL-STD-2084	Maintainability of Avionic and Electronic Systems and Equipment, General Requirements for
DOD-STD-2167	Defense System Software Development

HANDBOOKS

DOD-HDBK-263	Electrostatic Discharge Control Handbook for Protection of Electrical and Electronic Parts, Assemblies, and Equipment (excluding Electrically Initiated Explosive Devices) (Metric)

(Unless otherwise indicated, copies of federal and military specifications, standards, and handbooks are available from the Standardization Order Desk, 700 Robbins Ave, Bldg #4, Section D, Philadelphia, PA 19120-5094.)

2.1.2 Other Government publications. The following other Government publications form a part of this document to the extent specified herein. Unless otherwise specified, the issues are those cited in the solicitation.

PUBLICATIONS

NAVAL AIR SYSTEMS COMMAND

AD-1115	Electromagnetic Compatibility Design Guide for Avionics and Related Ground Support Equipment
AS-4613	Application and Derating, Requirements for
DARCOM-P-706-410	Engineering Design Handbook for Electromagnetic Compatibility
NAC TP-594	Interface Control Document, On Top Position Indicator (OTPI) Receiver AN/ARN-146
NAVMAT P-4855-1	Navy Power Supply Reliability Design and Manufacturer Guideline
NAVMAT P-9492	Navy Manufacturing and Screening Program Decreased Corporate Costs, Increased Fleet Readiness

Source: https://assist.dla.mil -- Downloaded: 2026-10-26T04:51Z
Check the source to verify that this is the current version before use.

(Application for copies should be obtained from the contracting activity or as directed by the contracting officer)

2.1.3 Order of precedence. In the event of a conflict between the text of this specification and the references cited herein, the text of this document takes precedence. Nothing in this document, however, supersedes applicable laws and regulations unless a specific exemption has been obtained.

3. REQUIREMENTS

3.1 First article. When specified (see 6.8), a sample(s) shall be subjected to first article inspection in accordance with 4.3.

3.2 Parts and materials. In the selection of parts, materials, processes and standards, the following shall govern:

a. Microelectric Technology shall be considered and microelectric items shall be in accordance with the requirements specified herein and in MIL-STD-454.

b. Microelectronic parts should be of the high reliability military grade type, in accordance with MIL-S-19500 and MIL-M-38510.

c. Other parts and materials requirements shall be in accordance with MIL-E-5400.

d. Nonrepairable subassemblies shall be used in accordance with MIL-STD-2084 and MIL-E-5400. The general size of the subassembly and the amount of circuitry to be included therein shall be approved by the procuring activity. Nonrepairable subassemblies must be reliable (see 6.6).

3.2.1 Nonstandard parts and materials approval. Approval for the use of nonstandard parts and materials (including electronic tubes, transistors and diodes) other than microelectronic devices shall be obtained in accordance with MIL-E-5400. Microelectronic devices shall be approved in accordance with MIL-STD-965.

3.2.2 Microelectronic modular assemblies. When used, microelectronic modular assemblies shall be in accordance with MIL-STD-965. Conformal coatings, encapsulants, embedments, or potting materials used with modular assemblies containing integrated circuits and discrete parts shall be easily removable without damage to the assembly.

3.2.3 Modules. The electronic portions of the receiving set shall be functionally modularized in accordance with MIL-STD-2084.

3.2.4 Parts derating. Electrical and electronic materials and parts shall be selected on a derated basis in accordance with AS-4613.

3.2.5 Keying. Modules and connectors shall be keyed to prevent the insertion of a connector or subassembly into an improper location within or on the receiving set.

Source: https://assist.dla.mil -- Downloaded: 2026-10-26T04:51Z
Check the source to verify that this is the current version before use.

3.2.6 Microelectronic devices. Maximum use shall be made of microelectronic devices in the design and construction of the receiving set. Conventional electronic devices shall be used only when it has been established that microelectronic devices are not suitable. All microelectronic devices used shall be in accordance with 3.2, to the greatest extent practical. Microelectronic devices selected for use in the receiving set which do not comply with 3.2 shall be subjected to the following requirements:

a. 100 percent Class B screening in accordance with the requirements of MIL-M-38510 and MIL-STD-883, Methods 5004 and 5008.

b. 100 percent lot control on all microcircuits, except that date code to contractor serial number traceability shall not be a requirement.

c. 100 percent burn-in at the service temperature extremes and 100 percent testing and rigorous inspection, including either temperature cycling or thermal shock (see a above).

3.2.7 Vacuum tubes and semiconductor devices. Vacuum tubes shall not be used. Semiconductor devices shall be used in accordance with the requirements specified in MIL-E-5400.

3.2.8 Nonmagnetic materials. Nonmagnetic materials shall be used for all parts of the equipment, except where magnetic materials are essential.

3.2.9 Magnesium and magnesium parts alloys. Magnesium and magnesium parts alloys shall not be used.

3.3 Design and construction. The receiving set shall be in accordance with MIL-E-5400 and MIL-STD-454 (less requirement 5) for design and construction, except as otherwise specified herein.

3.3.1 Total weight. The total weight of the receiving set, excluding cables, shall be consistent with good design and shall be not greater than six pounds.

3.3.2 Reliability development. The contractor shall establish and maintain an effective reliability program for the receiving set and when required by the contract specify the work in a Reliability Program Plan. The receiving set shall be planned, analyzed, designed, developed and tested in accordance with MIL-STD-781 and MIL-STD-785 to provide specified reliability. The contractor shall design and develop the receiving set through test, analyze and fix (TAAF) procedures in accordance with MIL-STD-781. Prior to the TAAF phase of the development, the contractor shall as a minimum define the environmental mission profile for the receiving set, develop reliability block diagrams and associated mathematical models, make reliability predictions for the components in accordance with MIL-STD-756, develop and perform in accordance with MIL-STD-1629 a Failure Mode Effects and Criticality Analysis (FMECA) on the components and conduct a Worst Case Analysis of all circuits and receiving set components to determine that the design will give proper response for all inputs at the specified limits of the environmental operating conditions. The contractor shall assign reliability values (MTBF hours) to the components and develop and maintain a reliability model to support this apportionment. The MTBF values of the receiving sets supplied or developed under this specification shall be apportioned to attain the receiving set reliability MTBF but in no case shall the MTBF be less than specified herein. The contractor shall monitor and control the work of subcontractors and suppliers to

Source: https://assist.dla.mil -- Downloaded: 202610-26T04:51Z
Check the source to verify that this is the current version before use.

assure the specified reliability of the receiving set. The contractor shall verify reliability design growth of the components by conducting a TAAF test program of sufficient duration to indicate instantaneous MTBF values as specified herein.

3.3.2.1 Operational stability. The receiving set shall operate with specified performance, continuously or intermittently, for a period of at least 3000 hours without the necessity for readjustment of any controls which are inaccessible to the operator during normal use.

3.3.2.2 Operating life. The receiving set shall have a total operating life of 20,000 hours with minimal servicing and replacement of parts.

3.3.2.3 Reliability in mean time between failure (MTBF). The receiving set, including any Built-in-Test provision, shall have 1500 hours as a lower test mean (operating) time between failures (θ_1). Receiving set reliability shall be demonstrated in accordance with 4.5.3.

3.3.2.4 Time totalizing meter. The receiving set chassis shall contain a time totalizing meter in accordance with MIL-M-7793.

3.3.2.5 Parts program (Task 207 of MIL-STD-785). Parts selected for use in the receiving set shall meet the following design, quality and rescreening requirements.

3.3.2.5.1 Stress derating analysis. The contractor shall perform stress derating analysis for the receiving set electrical/electronic parts to analytically verify their proper application and to estimate predicted reliability prior to proper start of development testing under extreme environmental and operating conditions. The anticipated functional and environmental stresses of the electronic equipment shall be specified and analyzed to verify that the parts are not overstressed for worst-case conditions. Derating levels shall be established to reduce the failure rates of the receiving set. The criteria shall establish margins of safety between part rated strengths and applied electrical stresses, considering worst-case environmental conditions, such as local maximum part ambient temperatures. The electronic component derating criteria shall be, as a minimum, AS-4613, Class A. Results of derating efforts, including analysis conducted, status of design compliance, problems encountered, and corrective action taken shall be presented at scheduled design reviews.

3.3.2.5.2 Parts quality. The development of parts, selection criteria, and application criteria shall include the following:

a. Microcircuits - MIL-M-38510 Class B.

b. Semiconductors - JANTX/JANTXV.

c. Resistors, ER (Established Reliability), and Class R Capacitors.

Other parts shall not be used without procuring activity approval. When a required microcircuit device is not identified on the QPL for MIL-M-38510, Product Assurance Level Class B, the device selected shall pass the MIL-STD-883, Method 5004, Class B screening requirements and Groups A and B lot quality conformance in accordance with MIL-STD-883, Method 5005, Class B, and shall have at least initial verifications of the Groups C and D requirements of MIL-STD-883, Method 5005, Class B. Hybrid microcircuit devices shall pass the screening and lot quality conformance requirements of MIL-STD-883, Method 5008.

Source: https://assist.dla.mil -- Downloaded: 2026-10-26T04:51Z
Check the source to verify that this is the current version before use.

3.3.2.5.3 Parts rescreening. Rescreen all microcircuits and semiconductors prior to installation as follows:

a. Screening tests shall include PIN (Particle Impact Noise) testing on all cavity semiconductors and microcircuits, and functional tests at minimum specified low, ambient, and maximum specified high temperatures.

b. Establish contractor in-house and vendor surveillance over the application of parts, materials, and processes during fabrication and assembly to ensure course of degradation and variability are isolated and controlled.

c. Maintain visibility over failed parts analysis to ensure proper corrective action has been implemented.

d. Determine and evaluate problem areas related to parts, materials, and processes, and initiate corrective action.

e. Review and analyze proposed changes in parts, materials, and processes resulting from test, manufacturing, and field experience.

3.3.3 Cabling and connections.

3.3.3.1 Cables and connections. The receiving set shall provide for the use of cables and connectors in accordance with MIL-E-5400.

3.3.3.2 Interconnection cabling. The receiving set shall be capable of specified operation using external wiring in accordance with MIL-W-5088. The external wiring shall be unshielded, except that a minimum number of wires may be shielded when demonstrated as necessary to meet interference control requirements and provided the assembly of the cable to its plugs may be easily accomplished. External cables and that portion of the connectors attached to the cables shall not be supplied as part of the receiving set. Polyvinylchloride (PVC) wire shall not be used.

3.3.3.3 Electrical connectors. Electrical connectors shall be in accordance with MIL-E-5400 except those external connectors used to interface units with aircraft wiring shall be in accordance with MIL-C-38999, Series III or Series IV (insert configurations shall be in accordance with MIL-STD-1353), MIL-C-81511.

3.3.4 Control panels. All rack or console mounted control panels shall be in accordance with MIL-C-6781 and MIL-C-81774. The configuration of all panels shall be approved by the procuring activity prior to production testing.

3.3.5 Interchangeability and replaceability. All components, subassemblies, and modules having the same part number shall be interchangeable and replaceable in accordance with MIL-I-8500. The contractor shall establish and maintain an effective configuration control program, and when specified in the contract, specify this work in accordance with MIL-STD-481 for hardware and DOD-STD-2167 and MIL-STD-483 for software.

3.3.6 Electromagnetic interference (EMI). The receiving set shall comply with the requirements of MIL-STD-461 for Class A1B equipment except as modified herein. These requirements shall be met for a set of modes representing all possible operation modes of the radio (see 4.3.1.3).

Source: https://assist.dla.mil -- Downloaded: 2026-10-26T04:51Z
Check the source to verify that this is the current version before use.

3.3.6.1 Design guidelines. The receiving set shall be designed in accordance with DARCOM-P-706-410, AD-1115, and MIL-STD-461.

3.3.6.2 Emission limits. The following MIL-STD-461, Part 2 limits shall apply:

a. CE01

b. CE03

c. CE06

d. CE07

e. RE01

f. RE02

3.3.6.3 Susceptibility limits. The following MIL-STD-461, Part 2 limits shall apply as modified:

a. CS01

b. CS02 except that this requirement shall apply to all neutrals and ground whether or not they are internally returned to chassis ground.

c. CS03

d. CS04

e. CS05

f. CS06

g. RS01

h. RS02

j. RS03. THe field levels and modulation shall be as specified in table I, RS03 Signals, and table II, High Level RS03 Signals. The susceptibility criteria of 3.3.6.4.2 does not apply to high level RS03 testing.

3.3.6.4 Susceptibility definition. The criteria specified in 3.3.6.4.1 and 3.3.6.4.2 shall be applied to determine whether the display is susceptible.

3.3.6.4.1 High level susceptibility. Any repeatable, non-commanded change in the control settings, panel settings, or modes of operation, of the radio shall be considered an indication of susceptibility. This shall apply for both transient and permanent changes. Also, the inability to change modes of operation while exposed to the susceptibility signal shall be considered an indication of susceptibility.

3.3.6.4.2 Low level susceptibility. The receiving set shall comply with the performance requirements of 3.4.

Source: https://assist.dla.mil -- Downloaded: 2026-10-26T04:51Z
Check the source to verify that this is the current version before use.

TABLE I. RS03 Signals.

Frequency (MHz)	Modulation	Power Density (mW/cm^2)		
		Peak[5]	Avg	v/m(Ref)[6]
0.014-850	AM 90% 1 kHz Sine Wave	-	.106	20
215-225[7]	PAM[1] 10 usec[2] PW[3] 200 pps[4]	.317	-	35
215-225	PAM 200 usec PW 200 pps	.317	-	35
400-450	PAM 15 usec PW 300 pps	.106	-	20
850-940[7]	PAM 2 usec PW 1000 pps	.106	-	20
850-940	PAM 125 usec PW 300 pps	.106	-	20
940-1215	PAM 10 usec PW 1000 pps	.106	-	20
1215-1365	PAM 10 usec PW 3000 pps	.106	-	20
1365-2900	PAM 1 usec PW 1000 pps	.106	-	20
2900-3100[7]	PAM 3 usec PW 1000 pps	1.6	-	78
2900-3100	PAM 70 usec PW 200 pps	1.6	-	78
3100-9000	PAM 1 usec PW 1000 pps	.106	-	20
9000-14000	PAM 0.2 usec PW 1000 pps	.106	-	20
14000-18000	PAM 0.2 usec PW 50000 pps	.106	-	20

1/ Pulse Amplitude Modulated.
2/ Microseconds.
3/ Pulse Width.
4/ Pulses Per Second.
5/ Peak Power Density for PAM signals is specified as the power density measured by an average power measuring device if the generator is run CW, such as, average power density within the pulse envelope.
6/ Volts Per Meter equivalent shown for reference only.
7/ Frequency ranges listed with more than one modulation shall be repeated as necessary to include all different modulations and field strength shown.

Source: https://assist.dla.mil -- Downloaded: 202640-26T04:51Z
Check the source to verify that this is the current version before use.

TABLE II. High level RS03 signals.

Frequency (MHz)	Modulation				Power Density (mW/cm2)[5]	v/m (ref)[6]
200	PAM[1]	10 usec[2]	PW[3]	200 pps[4]	10.6	200
220	PAM	200 usec	PW	200 pps	10.6	200
425	PAM	4 usec	PW	300 pps	10.6	200
900[7]	PAM	2 usec	PW	1000 pps	10.6	200
900	PAM	125 usec	PW	300 pps	10.6	200
1250	PAM	10 usec	PW	3000 pps	10.6	200
1350	PAM	5 usec	PW	3000 pps	10.6	200
3000[7]	PAM	3 usec	PW	1000 pps	10.6	200
3000	PAM	70 usec	PW	200 pps	10.6	200
3600	PAM	1 usec	PW	1000 pps	10.6	200
5600	PAM	1 usec	PW	600 pps	10.6	200
9100	PAM	0.5 usec	PW	1000 pps	10.6	200
9375	PAM	0.2 usec	PW	1000 pps	10.6	200
13900	PAM	0.2 usec	PW	50000 pps	10.6	200
15500	PAM	0.2 usec	PW	50000 pps	10.6	200
33200	PAM	0.2 usec	PW	2000 pps	0.106	20

1/ Pulse Amplitude Modulated.
2/ Microseconds.
3/ Pulse Width.
4/ Pulses Per Second.
5/ Peak Power Density for PAM signals is specified as the power density measured by an average power measuring device if the generator is run CW, such as, average power density within the pulse envelope.
6/ Volts Per Meter equivalent shown for reference only.
7/ Frequency ranges listed with more than one modulation shall be repeated as necessary to include all different modulations shown.

Source: https://assist.dla.mil -- Downloaded: 202610-26T04:51Z
Check the source to verify that this is the current version before use.

3.3.6.5 Ground plane interference. All receiver interface circuits shall be capable of specified performance when subjected to the following aircraft chassis noise between the interfacing WRAs:

a. Three volts RMS from 320 Hz to 500 Hz. (Shall be not greater than 150 ma RMS applied current.)

b. One volt RMS from 500Hz to 20MHz. (Shall be not greater than 150 ma RMS applied current for the frequencies between 500Hz and 50 KHz.) Power shall be not greater than one watt (CW) from a 50-ohm source for frequencies above 50 KHz.

c. Plus or minus eight volt pulses, 70 microseconds wide at 100 pps. Peak current spikes shall be not greater than 15 amperes.

3.3.6.6 Grounding requirements. The grounding system between WRAs of the receiving set shall be compatible and consistent with the aircraft grounding scheme specified in 3.3.6.6.1 through 3.3.6.6.3. All WRAs shall be grounded in such a manner as to prevent ground loops and common ground returns for signal and power circuits, provide shielding for signal circuits, minimize EMI, and to protect personnel from electrical hazards.

3.3.6.6.1 Chassis grounds. A ground wire of minimum length connected internally to the WRA chassis shall be provided at a pin on each primary power connector. No circuit shall be allowed to use this wire as its primary return.

3.3.6.6.2 Shield grounds. A separate connector pin shall be provided for each aircraft wire shield unless suitable grounding is provided in external cable backshells. The connector shield pins shall be grounded to the receiving set chassis inside the WRA adjacent to the connector mounting and by the shortest means practical.

3.3.6.6.3 Component grounds. All externally exposed metal parts, shields, control shafts, switch handles, connectors, and bushings shall be grounded to the chassis.

3.3.6.7 Bonding. A means of electrically bonding the receiving set to the aircraft chassis shall be provided and shall be in accordance with MIL-B-5087. Bonding resistance between the bonding surface and all parts of the receiving set case shall be not greater than 2.5 milliohms.

3.3.6.8 Shielding gaskets. Shielding gaskets and similar devices for RF applications shall avoid the use of materials which represent severe dissimilar metal combinations with the interface.

3.3.7 Maintainability. The maintainability program, built-in-test features, construction and packaging, provisions for test points and other maintainability parameters shall be in accordance with MIL-STD-2084 (see 4.3.1.5).

3.3.7.1 Compatibility with automatic test equipment (ATE). The receiving set shall be compatible with the ATE and shall be in accordance with MIL-STD-2076. When specified in the contract, ATE Test Programs shall be prepared in accordance with MIL-STD-2077. If a program exists for the receiving set, and changes to the receiving set are made which will affect the fault diagnosis procedure, changes to the existing Test Program shall be prepared as part of the receiving set changes in accordance with MIL-STD-2077.

Source: https://assist.dla.mil -- Downloaded: 202640-26T04:51Z
Check the source to verify that this is the current version before use.

3.3.7.2 BIT. In addition to the requirements of MIL-STD-2084, the receiver BIT shall provide self-test, a monitor capability, and fault isolation to the WRA. The BIT circuits shall be protected against internally induced transients and MIL-STD-704 type line transients or interrupts to prevent false BIT indications.

3.3.7.3 Operational readiness test. The BIT Operational Readiness Test shall be automatically initiated upon direction from the System Processor, and shall not exceed 1 minute to execute, excluding warm-up time. Provisions shall also be made for initiation of BIT via the unit's Control Panel in Control Panel applications via a momentary "BIT Initiate" switch. The equipment failure detection requirements of MIL-STD-2084 shall apply to all self-test features, and shall be such as to obviate the need for any auxiliary test device to perform either fault detection or isolation.

3.3.7.4 Status monitor. The requirement shall apply while the equipment is performing the specified mission function. Operator alerts and cues for reporting equipment fault status and for informing the pilots and sensor operators when action is required as a result of an equipment status change shall be provided. Status shall be provided on the data bus when requested. Receiver status summary information shall also be available to indicate equipment fault status and availability to the system. The on-line status monitoring function shall detect a minimum of the following percentages of the avionic subsystem non-intermittent component failures which affect system performance, excluding Built-In-Test (BIT) circuitry within 1 minute, excluding warm-up time, of their occurrence.

a. Receiver - 30 percent channelized.

b. Receiver - 98 percent unchannelized.

This requirement shall apply to avionic system functions in their operations (such as non-standby mode). Ninety-eight percent of all fault indications shall be the result of actual failures. Fault status state shall be reported via operator alerts, and through an update of an avionics equipment status summary data base.

3.3.7.5 Fault isolation. The receiver BIT shall isolate the faulty WRA in accordance with MIL-STD-2084 and 3.3.7.4 herein.

3.3.7.6 Maintenance provisions for field testing. Provisions for maintenance shall be in accordance with MIL-E-5400. Test points shall be provided on the front of the receiver to monitor significant voltages necessary to the maintenance cycle. The following test points shall be included:

a. Direction Finder Group Power Supply.

b. Receiver Power Supply.

c. Audio Output.

d. AGC.

e. Ground.

Additional internal test points as necessary shall be provided for ease of trouble analysis.

Source: https://assist.dla.mil -- Downloaded: 202610-26T04:51Z
Check the source to verify that this is the current version before use.

3.3.7.7 Mean-time-to-repair, organizational level. The mean-time-to-repair (MTTR) shall include the following functions:

a. Preparation time (turn-on, warm-up).

b. Fault detection.

c. Fault isolation to a WRA.

d. WRA replacement.

e. Alignment.

f. Checkout of repair.

MTTR is the summation of all maintenance downtimes listed above divided by the number of maintenance actions performed. For organizational maintenance, the MTTR shall be not greater than 30 minutes. The time to repair at the organizational level shall be not greater than 75 minutes at the 95th percentile.

3.3.7.8 Mean access and repair time, intermediate level. The mean access and repair time shall include the following functions:

a. SRA/chassis-mounted component access.

b. SRA/chassis-mounted component remove/replace.

c. WRA close-up.

Mean access and repair time is the summation of all maintenance downtimes listed above divided by the number of maintenance tasks performed. For intermediate maintenance, the mean access and repair time shall be not greater than 30 minutes. The time to obtain access and accomplish repair at the intermediate level shall be not greater than 75 minutes at the 95th percentile.

3.3.7.9 Protection against electrostatic discharge (ESD). The parts, SRA's, and items shall be so designed that the microelectric circuits are not susceptible to damage or upset from ESD due to handling by operating or maintenance personnel. DOD-HDBK-263 and DOD-STD-1686 shall be used as guides for providing this protection.

3.3.8 Nomenclature, nameplates and identification marking. Nomenclature and serial number assignment, nameplate approval and identification marking shall be in accordance with MIL-N-18307.

3.3.9 Standard conditions. The following conditions shall be used to establish normal performance characteristics under standard conditions and for making laboratory bench tests.

a.	Temperature	Room ambient ($25^{o}C \pm 10\ ^{o}C$)
b.	Altitude	Normal Ground
c.	Vibration	None
d.	Humidity	Room ambient up to 90% relative humidity

Source: https://assist.dla.mil -- Downloaded: 202610-26T04:51Z
Check the source to verify that this is the current version before use.

e.	Input power	115 ±1.0 Vac 400 Hz ±20 Hz and 27.5 ±1.5 Vdc at 20 watts
f.	Panel lighting	27.5 ±1.5 Vdc or Vac

3.3.10 Service conditions. The receiving set shall operate as specified under any of the environmental service conditions or reasonable combination of these conditions in accordance with MIL-E-5400 for Class 1B equipment except as specified herein.

3.3.10.1 Vibration. The receiving set shall operate as prescribed when subjected to the vibration requirements of Curve III of MIL-T-5422.

3.3.10.2 Altitude. The receiving set shall be capable of prescribed operation at altitudes up to 15,000 feet.

3.3.11 Warm-up time. The time required for the receiving set to warm up prior to operations shall be kept to a minimum and shall be not greater than 2 minutes under standard conditions and 5 minutes at extreme service conditions.

3.3.12 Input electrical power.

3.3.12.1 Operating power. The receiving set shall meet all applicable requirements of MIL-STD-704 and shall give specific performance when energized from the following power sources having characteristics and limits defined in MIL-STD-704.

a. AC Power (Single Phase), 115 V, Category B, 15 VA.

b. DC Power, 28 V, Category B, 20 Watts.

3.3.12.2 Lighting power. Input power for lighting will be 28 volts, 10 watts.

3.3.13 Output power. The receiving set shall be capable of continuously providing 225 Vdc ±10 percent at 30 milliamperes +20 percent tube plate output power for either the AM-608/ARA-25 or AM-1777/ARA-25A Electronic Control Amplifiers, except that the equipment shall provide 225 Vdc ±40 percent during the normal voltage transients of MIL-STD-704. With normal steady state voltage applied, this voltage shall contain no more than 0.50 volt ripple (RMS).

3.3.14 Internal adjustments. Adjustments shall be kept to a minimum and shall be easily accessible upon removal of the receiver cover. All adjustments shall be factory aligned and present prior to delivery so that the receiver is operating to the requirements herein. Alignment shall be stable for not less than 3000 normal operation hours.

3.3.15 Dust cover. An external dust cover shall be supplied which is readily removable without attachment to wiring or other obstructions. The cover shall provide sand and dust protection in accordance with MIL-E-5400.

3.3.16 Mount. A mounting plate providing for easy attachment and removal of the receiving set shall be supplied.

3.3.17 Safety. The receiving set shall be designed for safety in accordance with the requirements of MIL-E-5400; MIL-STD-454, Requirement 1; MIL-STD-882, Sections 4 and 5; and as specified in 3.3.17.1 through 3.3.17.3

Source: https://assist.dla.mil -- Downloaded: 2026-10-26T04:51Z
Check the source to verify that this is the current version before use.

3.3.17.1 Electromagnetic radiation exposure. The system design criteria shall include provisions for protection of personnel from RF electromagnetic, electrostatic and shock hazards in accordance with Requirement 1 of MIL-STD-454. Where possible, protection provisions shall be designed into associated subsystems/equipments. When protection by design is not technically feasible, adequate safety precautions shall be provided.

3.3.17.2 Toxicity. Personnel exposure to toxic air contaminants during system operation or maintenance shall be evaluated and minimized. In addition, any malfunction of the receiver including electrical fires shall not release into the cockpit or cabin area any gas, fumes, vapors or particulates which could incapacitate the crew and subsequently result in inability to successfully perform an emergency landing/ditching and egress.

3.3.17.3 Electronic equipment hazards. The new electronic equipment and modified portions of existing electronic equipment shall ensure freedom from personnel hazards in accordance with MIL-STS-454, Requirement 1.

3.4 Performance. Unless otherwise specified herein, values set forth to establish the requirements for prescribed performance apply to performance under both standard and extreme service conditions. When reduced performance under the extreme conditions is acceptable, tolerances or values setting forth acceptable variations from the performance under the standard conditions will be approved by the procuring activity.

3.4.1 Operation. The receiving set shall be compatible with, match impedance with, and accept the voltage and impedance output levels of the AM-3624/ARA-50 amplifier relay assembly and the AS-578/ARA-25, AS-578A/ARA-25 and OA-8697/ARD antenna assemblies. Antenna sharing capability shall be provided by an internal miniature coaxial relay. Relaying of tube plate power for the direction finder group shall be provided internally. Associated systems functions including AGC, sensitivity, audio and ADF shall also be switched as necessary. All receiver connectors as well as all circuit protective devices, if required, shall be located on the front of the unit. The receiver shall preamplify, convert, IF amplify, and amplitude detect the antenna output and provide acceptable voltage and impedance output levels to be compatible with the input requirements of the Electronic Control Amplifiers AM-1777/ARA-25A, AM-608/ARA-25, AM-3624/ARA-48, ARA-50, or OA-8697/ARD. The receiver and control box shall operate in conjunction with the AN/ARA-25, AN/ARA-50, or OA-8697/ARD Direction Finder Group to provide a sonobuoy homing capability as displayed on an appropriate bearing, heading indicator.

3.4.2 Frequency range. The receiving set shall operate over the range of frequencies from 136 to 174 MHz in such manner that any one of the 99 channels in table III may be selected.

3.4.3 Local oscillator. The local oscillator stability shall be such that any change in alignment, including reset error caused by cycling of the channel selector or drift caused by aging of component parts, shall be not more than ±10 kilohertz from the assigned channel frequency.

3.4.4 Overall gain stability. The gain stability of this receiving set shall be such that regenerative oscillatory conditions do not prevail for any value of carrier level at the antenna connector below 20,000 microvolts.

Source: https://assist.dla.mil -- Downloaded: 202610-26T04:51Z
Check the source to verify that this is the current version before use.

TABLE III. Frequency for sonobuoy uplink (136 to 173.5 MHz band).

New Allocation	New Allocation (Continued)	Existing Allocation
32 136.000	66 148.750	1 162.250
33 136.375	67 149.125	17 162.625
34 136.750	68 149.500	2 163.000
35 137.125	69 149.875	18 163.375
36 137.500	70 150.250	3 163.750
37 137.875	71 150.625	19 164.125
38 138.250	72 151.000	4 164.500
39 138.625	73 151.375	20 164.875
40 139.000	74 151.750	5 165.250
41 139.375	75 152.125	21 165.625
42 139.750	76 152.500	6 166.000
43 140.125	77 152.875	22 166.375
44 140.500	78 153.250	7 166.750
45 140.875	79 153.625	23 167.125
46 141.250	80 154.000	8 167.500
47 141.625	81 154.375	24 167.875
48 142.000	82 154.750	9 168.250
49 142.375	83 155.125	25 168.625
50 142.750	84 155.500	10 169.000
51 143.125	85 155.875	26 169.375
52 143.500	86 156.250	11 169.750
53 143.875	87 156.625	27 170.125
54 144.250	88 157.000	12 170.500
55 144.625	89 157.375	28 170.875
56 145.000	90 157.750	13 171.250
57 145.375	91 158.125	29 171.625
58 145.750	92 158.500	14 172.000
59 146.125	93 158.875	30 172.375
60 146.500	94 159.250	15 172.750
61 146.875	95 159.625	31 173.125
62 147.250	96 160.000	16 173.500
63 147.625	97 160.375	
64 148.000	98 160.750	
65 148.375	99 161.125	

3.4.5 Sensitivity. A 10-decibel (dB) signal-to-noise ratio shall exist at the audio output when a 2.5 microvolt signal, 30 percent modulated by a 100 Hz square wave, is applied at the antenna connector.

3.4.6 Single-channel bandwidth. For any given channel, the receiver minimum 6 dB bandwidth shall be not less than 190 KHz and not more than 260 KHz. The slope factor shall provide not less than 60 dB adjacent channel rejection.

3.4.7 IF rejection. The receiving set shall attenuate the IF center frequency at least 80 dB relative to any desired channel center frequency applied at the receiver input.

Source: https://assist.dla.mil -- Downloaded: 2026-10-26T04:51Z
Check the source to verify that this is the current version before use.

3.4.8 Image and other spurious responses. The receiving set shall attenuate image and other spurious responses external to the passband at least 60 dB compared to any desired channel center frequency except at one-half signal frequency the response shall be at least 45 dB down. There shall be no more than two spurious responses in any channel within the passband which are less than 60 dB down, and these shall be at least 50 dB down.

3.4.9 RF output protection. The receiving set shall be protected for input signals up to 500,000 microvolts at any frequency or combination of frequencies below 30,000 MHz.

3.4.10 Audio output characteristics.

3.4.10.1 Frequency response. The frequency response of the receiver shall be such that the audio output from 75-6000 Hz is flat within + 3 dB of the reference level output at 100 Hz when measured over an external 22,000 ohm resistive impedance.

3.4.10.2 Audio level. The root-mean-square (rms) level of audio output for 100 Hz shall be 0.5 volts.

3.4.10.3 Audio level range. When a 100 Hz, 30 percent square wave modulated carrier signal of 2.5 microvolts or greater is applied to the receiver, the amplitude detected and amplified audio output shall be adjustable from 0.25 volts rms minimum to 1.0 volt rms over an external 22,000 ohm resistive impedance.

3.4.11 Overall phase delay. Phase delay between the 100 Hz modulation entering the receiver and the 100 Hz demodulated audio signal used as the input to the Electronic Control Amplifiers of the AN/ARA-25 or AN/ARA-50 Direction Finder Group shall be 180 ±10 degrees. Phase delay between the 5.68 kHz demodulated audio signal used as the input to the Filter Detector of the OA-8697/ARD Direction Finder shall be 270 ±10 degrees.

3.4.12 RF bandpass filter. RF amplifier circuits shall feature a filter which shall have a center frequency of 155 MHz and a 1 dB passband from 136 MHz to 174 MHz.

3.4.13 Reduced size coaxial relay. Internal coaxial relay used for antenna sharing shall have not greater than 0.5 dB insertion loss and shall have not less than 40 dB isolation between open contacts measured at 225 megahertz.

3.4.14 Remote adequate signal strength indicator. The receiving set shall contain switching circuits and a threshold control that is adjustable and shall operate upon receiving an RF signal of -86 dBm at the receiver antenna connector. For OA-8697/ARD operation this level shall be changed to -66.5 dBm. It shall never operate in the absence of signal. When actuated, the switching circuits shall supply a ground to the external circuit. Depending upon circuit design, performance of the above function without the use of a threshold control is preferred.

3.4.15 Headphone output. A nominal audio output of four to six volts (rms) shall be supplied at 400 Hz. The output audio line shall be 600 ohms.

Source: https://assist.dla.mil -- Downloaded: 202610-26T04:51Z
Check the source to verify that this is the current version before use.

3.4.16 OA-8697/ARD compensation rogramming. Internal programming circuits shall be provided to establish the proper delay compensation inputs to the remote compensation programming terminals of the OA-8697/ARD. When the OTPI is deselected, programming inputs for the associated communications receiver shall be present within the OTPI. Compensation selection shall be fail safe in that if the OTPI is powered down, the appropriate communication receiver compensation shall be provided.

3.4.17 DF antenna selection. Terminations shall be provided at the OTPI connector such that when jumpered by the aircraft wiring associated with the OA-8697/ARD the OTPI receiver RF gain and detected baseband output characteristics required for use with this DF system will be automatically selected.

3.4.18 Radio interference suppression. Radio interference suppression shall be in accordance with MIL-STD-461.

3.5 Detail requirement. The receiving set shall consist of the items listed below and specified in 3.5.1 through 3.5.2.4.

a. Radio Receiver R-2330/ARN-146.

b. Control Set C-11699/ARN-146 (optional).

c. Mounting Plate MT-ARN/146 (optional).

3.5.1 Radio Receiver R-2330/ARN-146.

3.5.1.1 Radio receiver function. The radio receiver shall receive VHF sonobuoy signals and provide homing and On-Top Position Indicating (OTPI) information and provide this information to associated direction finding equipment, and shall operate in a 31 channel mode as a direct replacement for the R-1651/ARA OTPI Receiver. The radio receiver shall also operate in a 99 channel mode.

3.5.1.2 Radio receiver form factor. The radio receiver shall be enclosed in a single case and shall not be larger than 3.05 inches wide by 4.955 inches high by 5.30 inches long.

3.5.1.3 Radio receiver weight. The weight of the radio receiver shall be not greater than 4.75 pounds.

3.5.1.4 Radio receiver electrical connections. Connections to external circuits shall be provided for the Receiver Assembly R-2330/ARN-146 so as to be mateable with the aircraft wiring as specified in table IV. All connectors shall be qualified to the appropriate military specification. All MIL-C-39012 connectors shall be category C or D.

3.5.1.4.1 Radio receiver connector pin functions. The pin functions of the connectors shall be defined in NAC TP-594.

3.5.1.5 Radio receiver finish. The finish of the receiver shall be lusterless grey in accordance with MIL-E-5400 or MIL-F-7179 as applicable.

Source: https://assist.dla.mil -- Downloaded: 202640-26T04:51Z
Check the source to verify that this is the current version before use.

TABLE IV. Radio receiver electrical connections.

Function	Reference Designation	Type (Mates With)
RF Antenna Input	J-1	M39012/01-XXXX
RF Output for VHF/ADF	J-2	M39012/01-XXXX
Power/Control/ Output	J-3	MS3467E37-50S
	J-4	D38999/46WD18SN

3.5.2 Radio set control.

3.5.2.1 Radio set control functions. The control set shall provide the following functions:

a. Select any one of the 99 VHF sonobuoy channels.

b. Power on-off.

c. Adequate signal strength indication.

d. Initiate BIT.

3.5.2.2 Radio set control form factor/mounting. The control set size shall be as follows for console mounting:

a. Length - 5.00 inches including switches and connectors.

b. Length - 3.80 inches without switches and connectors.

c. Height - 2.25 inches.

d. Width - 5.75 inches from panel.

e. Width - 4.60 inches control box.

3.5.2.3 Radio set control weight. The control set weight shall be not greater than 1.25 pounds.

3.5.2.4 Radio set control electrical connections. Connections to external circuits shall be provided for the Control Set C-11699/ARN-146 so as to be mateable with the aircraft wiring as specified in table V.

3.5.2.4.1 Radio set control connector pin functions. The pin functions of the connectors shall be defined in NAC TP-594.

Source: https://assist.dla.mil -- Downloaded: 202640-26T04:51Z
Check the source to verify that this is the currentversion before use.

TABLE V. Radio set control electrical connections.

Function	Reference Designation	Type (Mates With)
Power/Control	J-1	MS3476W16-23S

3.5.2.5 Panel. The control panel shall conform to the requirements of MIL-C-6781, MIL-C-81774 and MIL-STD-1472. The configuration of all panels must be approved by the Procuring Activity prior to production testing. All panels shall provide IPL Red illumination within CIE chromaticity coordinates specified in MIL-C-25050.

3.5.2.6 Panel lighting. Control panel information panels shall supply red lighting.

3.5.2.7 Labeling. The labeling of displays and controls on panels shall be in accordance with the requirements of MIL-M-18012 and MIL-STD-783. The FUTURA semibold letter design with these heights (0.125, 0.156, and 0.188 inches) shall be selected from MIL-M-18012. A constant character-height to stroke-width ratio of 6.5 ±0.5 shall be used.

3.5.2.8 Panel finish. The finish of the control panel shall be lusterless black, color number 37038, in accordance with FED-STD-595 and MIL-C-6781.

3.5.3 Mounting base.

3.5.3.1 Mounting base function. The function of the mounting base shall be as specified in 3.3.16.

3.5.3.2 Mounting base form factor. The mounting base shall be not larger than 3.05 inches wide by 0.6 inches high by 5.7 inches long.

3.5.3.3 Mounting base weight. The weight of the mounting base shall be not greater than 0.6 pounds.

3.6 Workmanship. The receiving set shall be in accordance with MIL-E-5400 for workmanship and shall be inspected in accordance with Section 4. The receiving set shall be designed in accordance with MIL-STD-454 (less requirement 5), MIL-STD-275, MIL-P-55110, NAVMAT P-9492, NAVMAT P-4855-1, and MIL-STD-2000.

4. QUALITY ASSURANCE PROVISIONS

4.1 Responsibility for inspection. Unless otherwise specified in the contract or purchase order, the contractor is responsible for the performance of all inspection requirements (examination and tests) as specified herein. Except as otherwise specified in the contract or purchase order, the contractor may use his own or any other facilities suitable for the performance of the inspection requirements specified herein, unless disapproved by the Government. The Government reserves the right to perform any of the inspections set forth in this specification where such inspections are deemed necessary to assure supplies and services conform to prescribed requirements.

Source: https://assist.dla.mil -- Downloaded: 2026-10-26T04:51Z
Check the source to verify that this is the current version before use.

4.1.1 Responsibility for compliance. All items shall meet all requirements of Sections 3 and 5. The inspections set forth in this specification shall become a part of the contractor's overall inspection system or quality program. The absence of any inspection requirements in the specification shall not relieve the contractor of the responsibility of ensuring that all products or supplies submitted to the Government for acceptance comply with all requirements of the contract. Sampling inspection, as part of manufacturing operations, is an acceptable practice to ascertain conformance to requirements; however, this does not authorize submission of known defective material, either indicated or actual, nor does it commit the Government to accept defective material.

4.2 Classification of inspections. The inspections requirements specified herein are classified as follows:

a. First article inspection (see 4.3).

b. Initial production tests (see 4.4).

c. Acceptance tests (see 4.5).

4.3 First article inspection. First article inspection shall be conducted on a receiving set representative of the production receiving sets to be supplied under the contract. First article inspection shall be accomplished under the approved test procedure of 4.7.

4.3.1 Scope of tests. First article inspection shall include all tests deemed necessary by the procuring activity to determine that the receiving set meets all the requirements of this specification and the contract. First article tests shall include, but not be limited to:

a. Electrical (bench) tests (see 4.3.1.1).

b. Environmental tests (see 4.3.1.2).

c. Electromagnetic interference tests (see 4.3.1.3).

d. Reliability development tests (see 4.3.1.4).

e. Maintainability demonstration tests (see 4.3.1.5).

4.3.1.1 Electrical (bench) tests. Bench tests shall be conducted to demonstrate compliance with specified performance parameters. These tests shall be conducted under normal laboratory conditions. Inspection and examination to determine compliance with material and workmanship requirements shall be conducted at this time. High and low temperatures shall be used to evaluate temperature dependent parameters.

4.3.1.2 Environmental tests. Environmental tests shall be conducted in accordance with MIL-STD-810. Environmental tests shall include, but not be limited to, the following as specified in 4.3.1.2.1 through 4.3.1.2.8 (see 3.3.10).

4.3.1.2.1 Temperature/altitude test. A combined temperature/altitude test shall be conducted in accordance with MIL-STD-810, Methods 500, 501 and 502, to demonstrate compliance to requirements of MIL-E-5400 Class 1B equipment.

Source: https://assist.dla.mil -- Downloaded: 202640-26T04:51Z
Check the source to verify that this is the current version before use.

4.3.1.2.2 Shock test. Shock test shall be performed without vibration isolators (referred to as direct hard-mounting) and in accordance with MIL-STD-810, Method 5 and shall include the following test:

a. Basic Design Test - Procedure I, figure 516-1.

b. Crash Safety Test - Procedure V, figure 516-4.

c. Bench Handling Test - Procedure II.

4.3.1.2.3 Humidity test. A humidity test shall be performed in accordance with MIL-STD-810, Method 507.

4.3.1.2.4 Salt fog test. A salt fog test shall be performed in accordance with MIL-STD-810, Method 509.

4.3.1.2.5 Explosion-atmosphere test. An explosion-atmosphere test shall be performed in accordance with Procedure I of MIL-STD-810, Method 511. The upper operational altitude limit shall be 15,000 ft.

4.3.1.2.6 Temperature shock test. A temperature shock test shall be performed in accordance with MIL-STD-810, Method 503.

4.3.1.2.7 Rain test. Equipment mounted internally shall be subjected to a rain test in accordance with MIL-STD-810, Method 506, Procedure II.

4.3.1.2.8 Vibration. The test item shall be vibrated without vibration isolator (hard-mounted) and shall meet all performance requirements both during and after exposure to Curve III of MIL-T-5422 (see 3.3.10.1).

4.3.1.3 Electromagnetic interference (EMI) tests. Compliance with the requirements of 3.3.6 and its subparagraphs shall be demonstrated by tests in accordance with MIL-STD-462 Notice 2 as modified in 4.3.1.3.1 through 4.3.1.3.5.

4.3.1.3.1 EMI test plan. EMI tests shall be performed in accordance with an EMI test plan. The test plan shall specify all EMI tests required to demonstrate compliance with 3.3.6 and shall address all anomalies, inconsistencies, and omissions (if any). The test plan shall detail a combination of operating modes and frequencies to be tested. The test plan shall be acceptable to the procuring activity prior to the commencement of EMI testing.

4.3.1.3.2 CE01 test procedures. The requirements of 3.3.6.2.a shall be demonstrated in accordance with MIL-STD-462 Notice 2 Test Method CE01 for all power leads and MIL-STD-462 Notice 2 Test Method CE02 for all control and signal leads tested.

4.3.1.3.3 CE03 test procedures. The requirements of 3.3.6.2.b shall be demonstrated in accordance with MIL-STD-462 Notice 2 Test Method CE03 for all power leads and MIL-STD-462 Notice 2 Test Method CE04 for all control and signal leads tested.

4.3.1.3.4 CE07 test procedures. The contractor shall demonstrate compliance with the requirements of 3.3.6.2.d.

Source: https://assist.dla.mil -- Downloaded: 2026-10-26T04:51Z
Check the source to verify that this is the current version before use.

4.3.1.3.5 Ground plane interference. The requirements of 3.3.6.5 shall be demonstrated by test using the setup shown on figure 3.

4.3.1.4 Reliability development test (RDT). RDT shall be conducted in accordance with MIL-STD-781 with the following modifications:

a. The cumulative operating time shall be 6000 hrs.

b. No sample unit shall be operated for less than one-half the average operating time of all sample units on test.

c. Accept Criteria - The receiving set shall be judged to have satisfactorily completed the RDT when the total operating time has been completed. Acceptance shall be upon procuring activity approval of corrective action failures occurring during this test.

4.3.1.5 Maintainability demonstration test.

4.3.1.5.1 Selection of faults. Sample size and fault selection shall be in accordance with MIL-STD-471, Procedure 1, Method 9. Built-In-Test (BIT) failures if any shall be considered as candidates for fault selection. The procuring activity will select a random sample fault for each demonstration (see 3.3.7).

4.3.1.5.2 Accept/reject criteria. The accept/reject criteria for the demonstration of corrective maintenance times shall be in accordance with MIL-STD-471, Procedure 1, Method 9. An accept decision can only be made when an accept decision is made for all indices.

4.3.1.5.3 Technicians. The procuring activity reserves the right to provide and select the technicians to perform maintainability demonstration. The contractor shall provide adequate training to qualify the military technician(s) to perform the maintainability demonstration.

4.3.1.5.4 Technical documentation. Technical documentation to be used shall be limited to the receiving set and ground support equipment technical manuals.

4.3.1.5.5 Acceptance. The maintainability demonstration shall demonstrate that the system has met the accept criteria prior to the delivery of any system.

4.3.1.5.6 Rejection. If a rejection decision is reached in accordance with MIL-STD-471, Procedure 1, Method 9, the Procuring Activity shall be immediately notified. The contractor shall take the following action:

a. Develop a plan for design or material changes adequate to correct all deficiencies.

b. Upon acceptance of the plan by the procuring activity, perform corrective action and repeat the demonstration until an accept decision is reached.

4.3.2 First article approval. Approval of the first article sample will be by the procuring activity upon satisfactory completion of all tests. No production receiving sets shall be delivered prior to the approval of the first article sample. The first article sample shall not be considered as one of the receiving sets under

Source: https://assist.dla.mil -- Downloaded: 2026-10-26T04:51Z
Check the source to verify that this is the current version before use.

the contract. Should a failure occur during first article testing, the following action shall be taken:

a. Notify the procuring activity immediately.

b. Determine the cause of the failure.

c. Make available to the procuring activity for acceptance, the proposed corrective action.

d. Implement and verify corrective action and continue first article tests.

The procuring activity reserves the right to repeat any tests necessary to verify the corrective action.

4.3.3 Production receiving set. Receiving sets supplied under the contract shall in all respects, including design, construction, workmanship, performance and quality, be equal to the approved first article sample. Each receiving set shall be capable of successfully passing the same test as imposed on the first article sample. Evidence of non-compliance with the above shall constitute cause for rejection of receiving set under test and receiving set already accepted by the Government. The contractor shall make necessary corrections which are acceptable to the procuring activity.

4.4 Initial production tests. One of the first ten production receiving sets shall be selected and sent at the contractor's expense to a designated Government laboratory for test. This receiving set will be selected by the procuring activity after the receiving set has successfully passed all individual tests. No other tests shall be conducted on receiving set prior to starting the initial production tests. The first article sample shall not be selected for this test.

4.4.1 Scope of test. This receiving set may be subjected to any and all tests the procuring activity deems necessary to assure that the production receiving set is equal to the previously approved first article sample in design, construction, performance, and quality and that it meets all requirements of this specification and the contract.

4.4.2 Accessory material. In addition to the complete receiving set submitted for initial production tests, the contractor shall also submit such accessory material and include data necessary to test the receiving set.

4.4.3 Initial production sample approval. Approval of the initial production sample will be by the procuring activity upon satisfactory completion of all tests. Any design, material or performance defect made evident during these tests shall be corrected by the contractor to the satisfaction of the procuring activity. Failure of the initial production sample to pass any of the tests shall be cause for deliveries of the receiving set under the contract to cease until proper corrective action is approved and accomplished. Corrective action shall also be accomplished on receiving sets previously accepted when requested by the procuring activity.

4.5 Acceptance test. The contractor shall furnish all samples and shall be responsible for accomplishing the acceptance tests. All inspection and testing will be under the supervision of the Government inspector. Acceptance tests shall consist of the following:

Source: https://assist.dla.mil -- Downloaded: 2026-10-26T04:51Z
Check the source to verify that this is the current version before use.

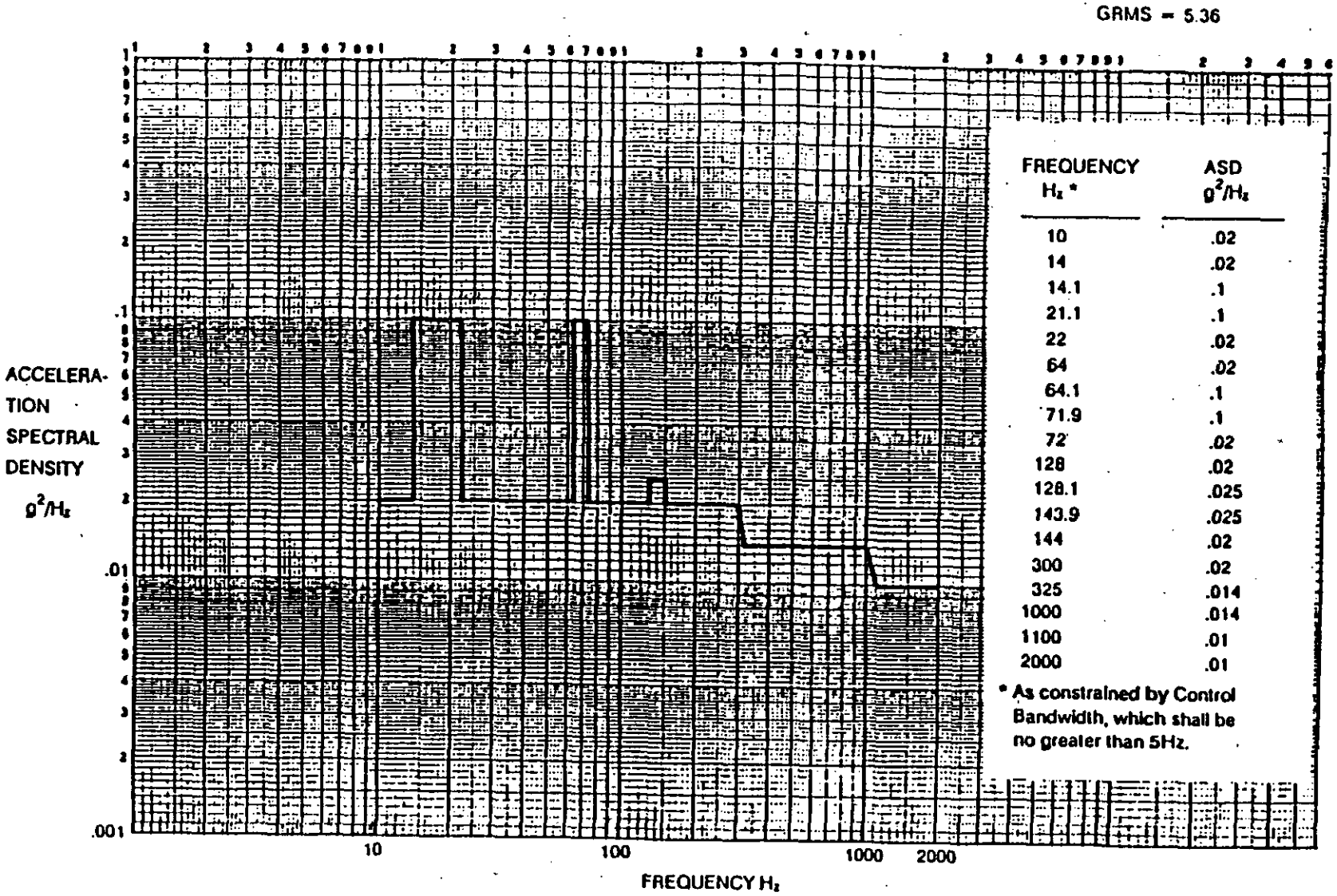


FIGURE 1. Random vibration test requirements.

Source: https://assist.dla.mil -- Downloaded: 202610-26T04:51Z
Check the source to verify that this is the current version before use.

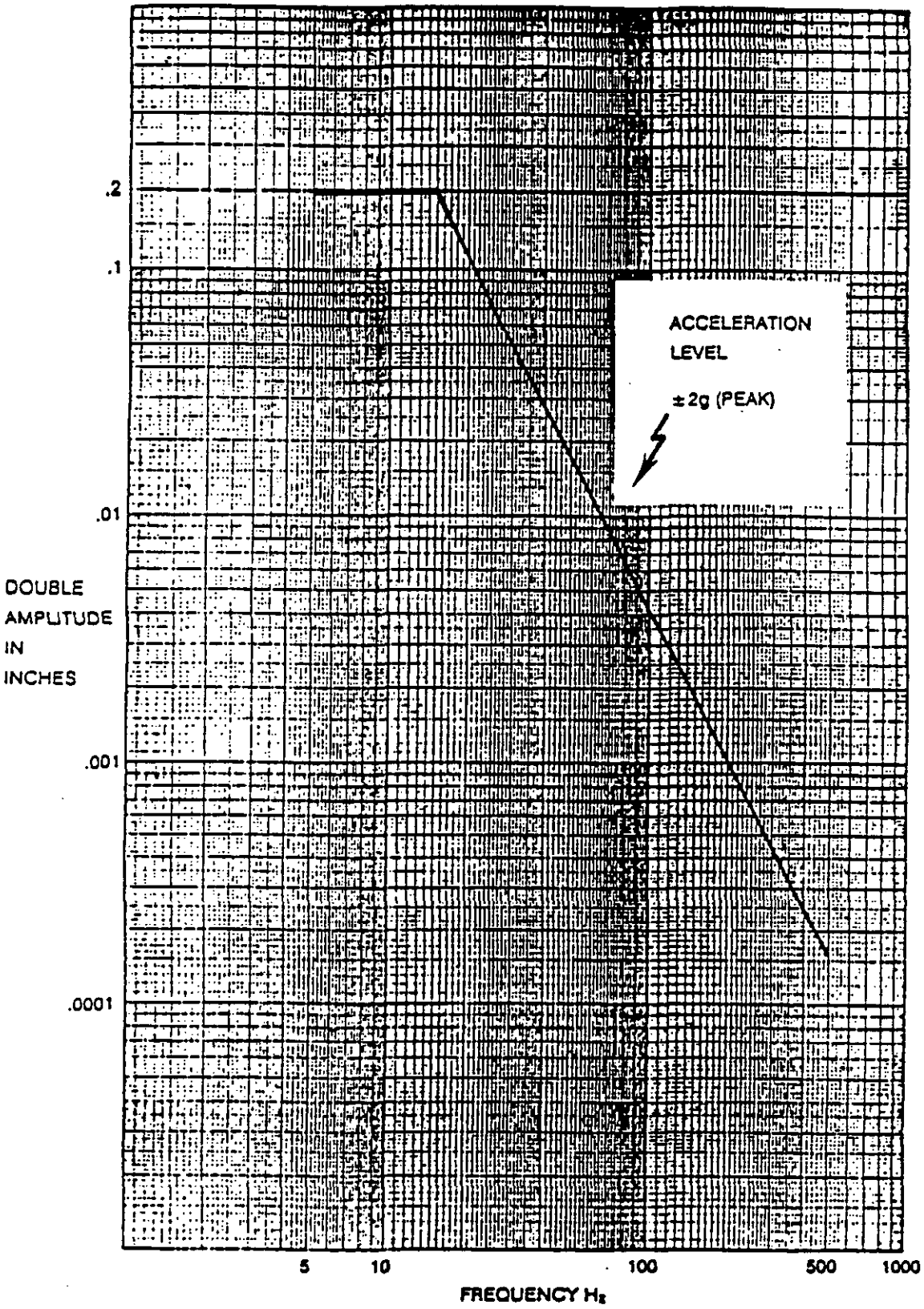
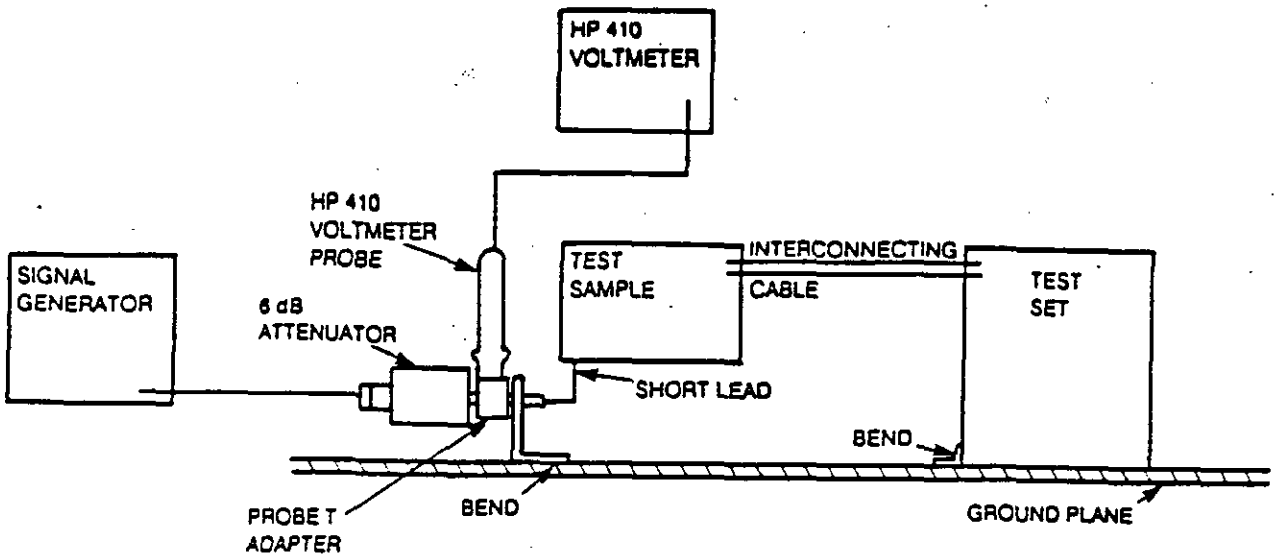
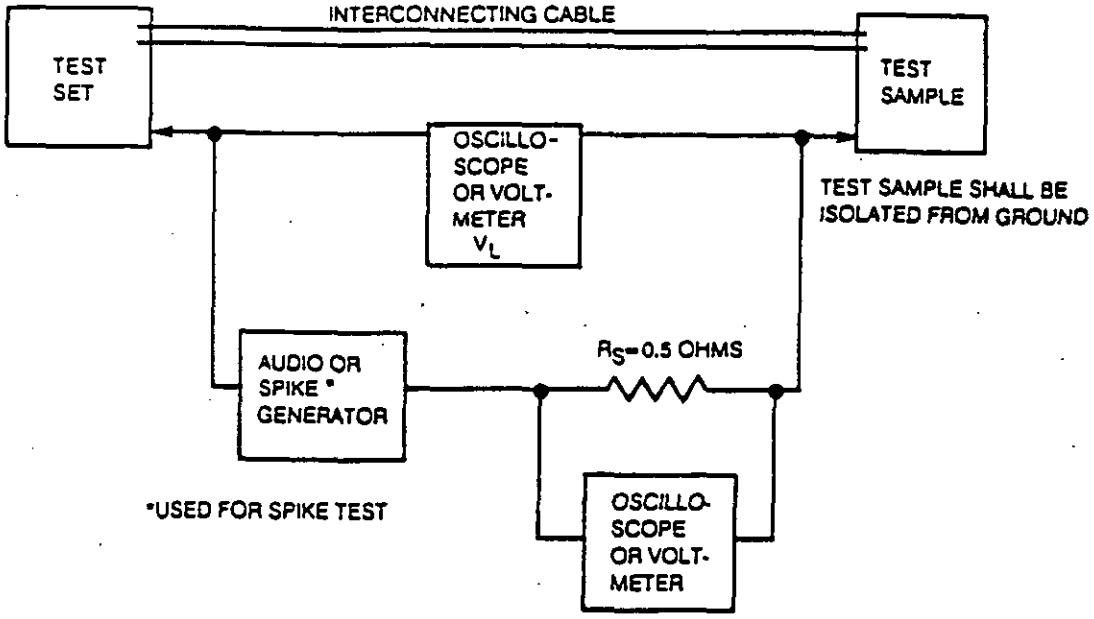


FIGURE 2. Sinusoidal vibration test requirements.

Source: https://assist.dla.mil -- Downloaded: 202610-26T04:51Z
Check the source to verify that this is the current version before use.



GROUND PLANE INTERFERENCE TEST SETUP (0.05 TO 100 MH_z)



GROUND PLANE INTERFERENCE TEST SETUP SPIKE TEST AND 320 H_z TO 50 KH_z TEST

FIGURE 3. Ground plan test requirements.

Source: https://assist.dla.mil -- Downloaded: 202610-26T04:51Z
Check the source to verify that this is the current version before use.

a. Individual tests (see 4.5.1).

b. Sampling tests (see 4.5.2).

c. Production reliability acceptance tests (see 4.5.3).

d. Special tests (see 4.5.4).

4.5.1 Individual tests. Each receiving set submitted for acceptance shall be subjected to the individual tests. These tests shall be adequate to determine compliance with the requirements of material, workmanship, operational adequacy and reliability. As a minimum, each receiving set accepted shall have passed the following tests:

a. Examination of product (see 4.5.1.1).

b. Operational tests (see 4.5.1.2).

c. Manufacturing run in tests (see 4.5.1.3).

4.5.1.1 Examination of product. Each receiving set shall be examined carefully to determine that the material and workmanship requirements have been met, and the physical characteristics of the receiving set agree with the engineering drawings acceptable to the procuring activity.

4.5.1.2 Operational test. Each receiving set shall be operated long enough to permit the receiving set temperature to stabilize and to check sufficient characteristics and record adequate data to assure specified receiving set operation.

4.5.1.3 Manufacturing run in test. An operational test shall be performed before and after manufacturing screening. Each receiving set shall be subjected to manufacturing screening of the following:

a. Random vibration (see 4.5.1.3.1).

b. Temperature cycling (see 4.5.1.3.2).

4.5.1.3.1 Random vibration. The receiving set shall be subjected to 6 G rms random vibration in accordance with figure 2 of MIL-STD-781. The receiving set shall be vibrated without vibration isolators with operating power applied. The direction of vibration shall be vertical to the normal mounting plane for 5 minutes and for 5 minutes perpendicular to the plane of the majority of the printed circuit cards. When it is not feasible to vibrate the receiving set in two directions, the vibration shall be for 10 minutes perpendicular to the plane of the majority of the printed circuit boards.

4.5.1.3.2 Temperature cycling. The receiving set shall be subjected to 10 thermal cycles between -54°C and +55°C. The receiving set shall be turned on and operating during the heating period and off during the cooling period. The duration of exposure at the high and low temperatures shall be 10 minutes or sufficiently long to allow the total mass of the receiving set to reach the specified temperature, whichever is greater. The temperature chamber shall have sufficient thermal capacity to provide a temperature change of at least 5°C per minute.

Source: https://assist.dla.mil -- Downloaded: 2026-10-26T04:51Z
Check the source to verify that this is the current version before use.

4.5.2 Sampling tests. Receiving sets selected for sampling tests shall first have passed the individual tests. Receiving sets will be selected for sampling tests by the Government inspector in accordance with table VI.

TABLE VI. Sampling tests sample selections.

Quantity of Equipments Offered for Acceptance	Quantity to be Selected for Sampling Test
First 10 1/	1
First 50	1
First 75	1
First 100	1
one for each additional 200 or fraction thereof	

1/ One out of first ten need not be selected and tested if initial production tests are conducted. Sampling Tests are not required when Reliability assurance tests are conducted.

4.5.2.1 Scope of test. As a minimum, each receiving set selected for sampling tests shall be subjected to the following tests under the cognizance of the Government inspector:

a. Complete operational test at ambient room conditions, making all necessary measurements to assure that all applicable requirements of this specification have been met.

b. Operational test at certain environmental conditions. The conditions should be based on results of the first article, initial production, individual, and special tests.

4.5.3 Production reliability acceptance tests. Production reliability acceptance tests shall be conducted in accordance with MIL-STD-781. Tests as required by both the Qualification Phase and the Production Acceptance (Sampling) Phase shall be conducted. Classification of failure shall be in accordance with MIL-STD-781 and MIL-STD-2074 (see 3.3.2.3).

4.5.3.1 Qualification phase. Prior to the acceptance of receiving sets under the contract or purchase order, not less than three receiving sets shall be tested in accordance with MIL-STD-781, for category 5 equipment. The applied vibration shall be swept sinusoidal from 5 to 500 Hz with the acceleration level shown on figure 2. For the Qualification Phase, Test Plan III shall be used. The accept-reject criteria for Test Plan III shall be used.

4.5.3.2 Reliability production acceptance phase tests. The receiving set, throughout production, shall be subjected to the All Equipment Test in accordance with MIL-STD-781 (as modified herein) for category 5 equipment. Each receiving set produced except those submitted for the Reliability Qualification Test, shall be tested for 50 hours. The final 25 hours shall be failure free. Manufacturing run in test shall be performed prior to the 50 hour test. To determine whether the MTBF is being met at any time during the contract the operating test hours and the

Source: https://assist.dla.mil -- Downloaded: 2026-10-26T04:51Z
Check the source to verify that this is the current version before use.

failures thereon shall be totaled and the results compared with the reject line of Test Plan III of MIL-STD-781. (Extend the line as necessary to accommodate the data). These totals shall accumulate so that at any one time the experience from the beginning of the contract is included. At any time that the current totals of test hours and test failures plotted on Test Plan III curves show a reject situation, the procuring activity shall be notified. The procuring activity reserves the right to stop the acceptance of the receiving sets at any time that a reject situation exists pending a review of the contractor's efforts to improve the receiving set, the receiving set parts, and the receiving set workmanship so that the entire compilation shows other than a reject decision.

4.5.3.3 Test details. The test details such as the length of the test cycle, and the length of the heat portion of the cycle, the performance characteristics to be measured, special failure criteria, and preventive maintenance to be allowed during the test shall be part of the test procedures to be reviewed by the procuring activity prior to the beginning of the Qualification Test Phase of the Reliability Assurance Tests.

4.5.3.4 Burn-in requirement for spares. The reliability testing procedures herein shall be applied to all assemblies (SRA and sub-SRAs) that are procured separately from the receiving set.

a. As a minimum, in-process burn-in, screening, and other individual testing of separately procured assemblies shall be equal to that conducted on production assemblies. Following the burn-in, a complete functional performance test shall be conducted on each assembly offered for acceptance.

b. Included in this testing shall be at least 75 hours power-ON operation (signal voltage not required) of each assembly under the environmental test conditions specified in 4.5.1.3. Power supply assembly burn-in shall be conducted with the power supply loaded at the rated full load of the receiving set.

c. The tests specified herein shall be conducted by or for the supplier of the assemblies. Assemblies shall not be accepted by contractors for delivery to the procuring activity or a Government activity, without objective evidence of having passed the above and other required tests. These test requirements shall be included in the manufacturing drawing, specification, or source controlled drawing for the assemblies.

4.5.4 Special tests. Special tests shall be conducted for the purpose of checking the effect of any design or material change on the performance of the receiving set and to assure adequate quality control. The receiving set selected for special tests may be selected from receiving sets previously subjected to the sampling or reliability assurance tests.

4.5.4.1 Special test schedule. Selection of receiving sets for special tests shall be made as follows:

a. On an early receiving set after an engineering or material change.

b. Whenever failure reports or other information indicate that additional tests are required, as determined by the procuring activity.

Source: https://assist.dla.mil -- Downloaded: 2026-10-26T04:51Z
Check the source to verify that this is the current version before use.

4.5.4.2 Scope of tests. Special tests shall consist of such tests as are acceptable to the procuring activity. The procedures previously approved for the first article tests shall be used where applicable.

4.5.5 Equipment failure. Should a failure occur during either the sampling, reliability assurance or special tests, the following action shall be taken:

a. Determine the cause of failure.

b. Determine if the failure is an isolated or design defect.

c. Include for procuring activity acceptance, proposed corrective action intended to reduce the possibility of the same failure(s) occurring in future tests.

4.6 Failure report. In the event of a failure, the Government inspector shall be notified immediately.

4.7 Test procedures. The right is reserved by the procuring activity or the Government inspector to modify the tests or require any additional tests deemed necessary to determine compliance with the requirements of this specification and the contract. MIL-T-18303 shall be used as a guide for preparation of test procedures. When approved test procedures are available from previous contracts, such procedures will be provided and may be used when their use is approved by the procuring activity. However, the right is reserved by the procuring activity to require modification of such procedures, including additional tests, when deemed necessary.

4.8 Presubmission testing. No items, part or complete receiving set shall be submitted by the contractor until it has been previously tested and inspected by the contractor and found to comply with all applicable requirements of this specification and the contract.

5. PACKAGING

5.1 General. All major units and parts of the receiving set shall be preserved, packed, and marked for the level of shipment specified in the contract in accordance with MIL-E-17555 and MIL-STD-794. In the event the receiving set is not covered in MIL-E-17555, the method of preservation for level A shall be determined in accordance with the selection chart in Appendix D of MIL-STD-794.

6. NOTES

(This section contains information of a general or explanatory nature that may be helpful, but is not mandatory.)

6.1 Intended use. This Receiving Set AN/ARN-146 is intended for use in ASW Communications on Naval aircraft.

6.2 Acquisition requirements. Acquisition documents should specify the following:

a. Title, number, and date of this specification.

b. Selection of applicable levels of testing (see Section 4).

Source: https://assist.dla.mil -- Downloaded: 2026-10-26T04:51Z
Check the source to verify that this is the current version before use.

c. Corrective action on previously accepted requirements, if required (see 4.4.3).

d. Selection of applicable levels of packaging and packing (see 5.1).

6.3 Consideration of data requirements. The following data requirements should be considered when this specification is applied on a contract. The applicable Data Item Description (DID's) should be reviewed in conjunction with the specific acquisition to ensure that only essential data are requested/provided and that the DID's are tailored to reflect the requirements of the specific acquisition. To ensure correct contractual application of the data requirements, a Contract Data Requirements List (DD Form 1423) must be prepared to obtain the data, where DOD FAR Supplement 27.475-1 exempts the requirement for a DD Form 1423.

Reference Paragraph	DID Number	DID Title	Suggested Tailoring
3.2.1	DI-E-7028A	Nonstandard Part Approval Requests	
3.2.2	UDI-E-21337	Microelectronic Devices	
3.3.2	DI-R-7079	Plan, Reliability Program	
3.3.4	UDI-E-21339	Drawing Requirements for Rack or Console	
3.3.5	DI-MCCR-80009	Configuration Control Program	
3.3.5	DI-E-2038	Change Proposals	
3.3.5	UDI-E-21335A	Avionic Change Directives	
3.3.8	DI-E7194	Request for Nomenclature	
4.3	UDI-R-21138	Report, Environment	
4.3.1.3.1	DI-EMCS-80201	Electromagnetic Interference Test Plan	
4.3.1.5.6, 4.5.5	DI-R-1704	Corrective Action Reporting Plan	
4.5, 4.5.1.3, 4.5.4.2, 4.7	UDI-T-21347	Test Procedures	
4.5, 4.5.3.2	DI-RELI-80252	Reliability Test Reports	
4.5.1.1	DI-E-7031	Engineering Drawings and Associated Lists	

6.4 Precedence of documents. When the requirements of the contract, this specification, or applicable referenced specifications are in conflict, the following precedence should apply:

Source: https://assist.dla.mil -- Downloaded: 202640-26T04:51Z
Check the source to verify that this is the current version before use.

a. Contract - The contract should have precedence over any specification.

b. This Specification - This specification should have precedence over all applicable subsidiary specifications. Any deviation from this specification, or from subsidiary specifications where applicable, should be specifically approved in writing by the procuring activity.

c. Referenced Specifications - Any referenced specification (Section 2.) should have precedence over all applicable subsidiary specifications referenced therein. All referenced specifications should apply to the extent specified.

6.5 Performance objectives. Minimum size and weight, simplicity of operation, ease of maintenance, and an improvement in the performance and reliability of the specific functions beyond the requirements of this specification are objectives which will be considered in the production of this receiving set. If simplicity of design, performance, ease of maintenance or reliability will result from the use of materials, parts and processes other than those specified in MIL-E-5400, it is desired that their use be investigated. When investigation shows advantages can be realized, a request for approval should be submitted to the procuring activity for consideration. Each request should be accompanied by complete supporting information.

(The above DID's were those cleared as of the date of this specification. The current issue of DOD 5010.12L, Acquisition Management Systems and Data Requirements Control List (AMSDL), must be researched to ensure that only current, cleared DID's are cited on the DD Form 1423.)

6.6 Nonrepairable subassemblies. As a general rule, nonrepairable subassemblies should be encapsulated or hermetically-sealed. The number of connections internal to the subassembly should be held to a minimum. Detail parts tolerances and ratings should be so selected that the life of the subassembly is greater than that of a similar repairable one. With few exceptions, nonrepairable subassemblies should evidence a Mean-Time-To-Failure greater than 5,000 hours, and for many applications this figure should be nearer 50,000 hours.

6.7 Type designations. The type designation may be modified by the procuring activity upon application by the contractor for assignment of nomenclature. The correct type number should be used on nameplates, shipping records and instruction books, as applicable.

6.8 First article. When first article inspection is required, the contracting officer should provide specific guidance to offerors whether the item(s) should be a first article sample, a first production item, or a standard production item from the contractor's current inventory and the number of items to be tested as specified in 4.4. The contracting officer should include specific instructions in acquisition documents regarding arrangements for examinations, approval of first article test results and disposition of first articles. Invitations for bids should provide that the Government reserves the right to waive the requirement for samples for first article inspection to those bidders offering a product which has been previously acquired or tested by the Government, and that bidders offering such products, who wish to rely on such production or test, must furnish evidence with the bid that which prior Government approval is presently appropriate for the pending contract.

Source: https://assist.dla.mil -- Downloaded: 202610-26T04:51Z
Check the source to verify that this is the current version before use.

6.9 Associated equipment. The receiving set covered by this specification should operate satisfactorily with the associated equipment listed in table VII which should not be supplied as part of the receiving set.

TABLE VII. Associated equipment.

Equipment	Specification
AN/SSQ-41 Sonobuoy	MIL-S-22793
AN/SSQ-36 Sonobuoy	MIL-S-22356
AN/SSQ-47 Sonobuoy	MIL-S-81349
AN/SSQ-50 Sonobuoy	MIL-S-81666
AN/SSQ-57 Sonobuoy	MIL-S-81478
AN/SSQ-53 Sonobuoy	MIL-S-81487
AN/SSQ-48 Sonobuoy	MIL-S-24020
AN/SSQ-62 Sonobuoy	MIL-S-81895
AN/SSQ-83 Sonobuoy	MIL-S-22793
AN/SSQ-86(XN-1) Sonobuoy	MIL-S-85020
AN/SSQ-71 Sonobuoy	MIL-S-85020
AN/SQQ-77 Sonobuoy	MIL-S-85313
OA-8697/ARD	---
AN/ARA-25 Direction Finder Group	MIL-D-7033
AN/ARA-50 Direction Finder Group	MIL-D-38402
AN/ARR-84 Sonobuoy Receiver	MIL-R-85689
R-1651/ARA, Receiver, Radio	MIL-R-81680
Bathythermograph	MIL-R-81355

6.10 Subject term (key word) listing.

Airborne ASW Sonobuoy Receiver
ASW Communications
Direction finder equipment
On-Top Position Indicator
Radio receiver
Radio set control

Preparing Activity
Navy-AS
(Project 5826-N213)

Source: https://assist.dla.mil -- Downloaded: 2026-10-26T04:51Z
Check the source to verify that this is the current version before use.

*U.S. GOVERNMENT PRINTING OFFICE: 1991-554-034/50256

STANDARDIZATION DOCUMENT IMPROVEMENT PROPOSAL

INSTRUCTIONS

1. The preparing activity must complete blocks 1, 2, 3, and 8. In block 1, both the document number and revision letter should be given.
2. The submitter of this form must complete blocks 4, 5, 6, and 7.
3. The preparing activity must provide a reply within 30 days from receipt of the form.

NOTE: This form may not be used to request copies of documents, nor to request waivers, or clarification of requirements on current contracts. Comments submitted on this form do not constitute or imply authorization to waive any portion of the referenced document(s) or to amend contractual requirements.

I RECOMMEND A CHANGE:	1. DOCUMENT NUMBER MIL-R-85690B	2. DOCUMENT DATE (YYMMDD) January 18, 1991

3. DOCUMENT TITLE
Radio Receiving Set AN/ARN-146

4. NATURE OF CHANGE (Identify paragraph number and include proposed rewrite, if possible. Attach extra sheets as needed.)

5. REASON FOR RECOMMENDATION

6. SUBMITTER

a. NAME (Last, First, Middle Initial)	b. ORGANIZATION	
c. ADDRESS (Include Zip Code)	d. TELEPHONE (Include Area Code) (1) Commercial (2) AUTOVON (If applicable)	7. DATE SUBMITTED (YYMMDD)

8. PREPARING ACTIVITY

a. NAME	b. TELEPHONE (Include Area Code) (1) Commercial (2) AUTOVON
c. ADDRESS (Include Zip Code)	IF YOU DO NOT RECEIVE A REPLY WITHIN 45 DAYS, CONTACT: Defense Quality and Standardization Office 5203 Leesburg Pike, Suite 1403, Falls Church, VA 22041-3466 Telephone (703) 756-2340 AUTOVON 289-2340

Source: https://assist.dla.mil -- Downloaded: 202610-26T04:51Z
Check the source to verify that this is the current version before use.